

Embedded Systems: Fast Track for Smart Brains

Professionals trained in embedded systems technologies are a rare commodity in the recruitment marketplace. Considering the vast scope of the field, ranging from telecom to consumer electronics to aerospace, the demand for embedded systems engineers for product development and application will continue to grow in the years to come



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Most of the electronic devices around us contain a processor with some embedded software. Unlike desktop computers, which can be programmed to do any general-purpose task, these embedded systems do highly focused and specific jobs.

The market for embedded systems is quite big. Increasing sales of consumer electronics goods, telecom/networking market and growth in the use of portable/wireless products globally are driving the growth of the semiconductor design industry.

If you have a creative and innovative bent of mind, a career in embedded systems can be extremely

rewarding. The industry offers a variety of jobs in the areas of chip design, applications engineering, product and test engineering, systems design and embedded software development. The positions in these job families range from the role of a technical individual contributor to technical lead position to an engineering management role.

Lots of embedded systems work—in particular, embedded software development—is being done in India. The embedded software industry is expanding as third-party service providers are moving up the value chain. Captives largely work on the hardware abstraction layer and the device driver layer. While many captives outsource application-related work to third-party service providers and Indian design companies to shorten their time-to-market, others prefer to grow their domain-specific embedded application expertise in-house.

Embedded system fundamentals

An embedded system consists of hardware and software. When chips are built into systems and software is loaded on that for a particular functionality, it becomes an embedded system. VLSI is used to design integrated circuits by combining millions and billions of transistors into a single chip, and involves various design activities ranging from specification to tape-out of the chip. VLSI results in reduced sizes and costs for embedded systems. Instead of using separate chips for the processor, memory and associate digital circuitry,